

ECON 695 — Midterm 1 Practice Exam

Time: 75 minutes

Closed-notes except formula sheet

Question 1. Conditional Expectation and Dummy Variables (15 points)

Suppose y_i denotes hourly wage and D_i indicates college completion ($D_i = 1$ if college, 0 otherwise).

- (a) Write the population conditional expectation function $E[y_i | D_i]$.
- (b) Show that the best linear predictor of y_i on $(1, D_i)$ is equivalent to the conditional means of y_i across the two groups.
- (c) Interpret α and β in the model $y_i = \alpha + \beta D_i + u_i$.
- (d) Extend this to three mutually exclusive education categories (less than HS, HS, College+). Write the regression specification and interpret each coefficient.

Question 2. Frisch–Waugh and Orthogonality (30 points)

You observe (y_i, x_{1i}, x_{2i}) with sample means 0. Consider:

$$y_i = \beta_1 x_{1i} + \beta_2 x_{2i} + u_i.$$

- (a) State the **Frisch–Waugh Theorem** for $\hat{\beta}_1$.
- (b) Derive $\hat{\beta}_1 = \frac{\sum_i \hat{\xi}_i y_i}{\sum_i \hat{\xi}_i^2}$, where $\hat{\xi}_i$ is the residual from the auxiliary regression $x_{1i} = \pi x_{2i} + \hat{\xi}_i$.
- (c) Suppose x_{1i} and x_{2i} are orthogonal in the sample (i.e., $\sum_i x_{1i} x_{2i} = 0$). Show that including or excluding x_{2i} does not affect $\hat{\beta}_1$.

- (d) (**Double FWL**) Let x_{1i} be endogenous and z_i an instrument. Using Frisch–Waugh twice, derive the 2SLS slope formula

$$\hat{\beta}_1^{2SLS} = \frac{\sum_i \tilde{z}_i \tilde{y}_i}{\sum_i \tilde{z}_i \tilde{x}_{1i}},$$

where tildes denote residuals after partialling out control x_{2i} .

Question 3. Oaxaca–Blinder Decomposition (30 points)

You study wage differences between two groups, a (private) and b (public). Let $y_i = \log(\text{wage}_i)$ and x_i represent education.

Group-specific regressions:

$$\begin{aligned} y_i &= (x_i)' \beta^a + u_i, & i \in a, \\ y_i &= (x_i)' \beta^b + u_i, & i \in b. \end{aligned}$$

- (a) Define the counterfactual mean for group b if it had group a 's education distribution but kept its own returns β^b .
- (b) Derive the Oaxaca–Blinder decomposition:

$$\bar{y}_b - \bar{y}_a = (\bar{x}_b - \bar{x}_a)' \beta^b + (\beta^b - \beta^a)' \bar{x}_a.$$

Label each component (composition vs. returns effect).

- (c) Using a simple “returns to education” example, interpret each term economically.
- (d) Suppose education is discrete (HS, BA, Grad). Write the counterfactual mean in terms of group-specific dummy variables D_g and group means y_g^b .

Checklist of Key Ideas for Review

- Orthogonality condition: $E[x_j u] = 0$ implies unbiased OLS.
- Frisch–Waugh: $\beta_j = \frac{(y, \xi_j)}{(\xi_j)}$, where ξ_j residualizes x_j on others.
- CEF for categorical regressors: $E^*[y|D] = E[y|D = 0] + D(E[y|D = 1] - E[y|D = 0])$.
- Oaxaca decomposition: “Between” effect = composition, “Within” effect = returns.
- Double Frisch–Waugh = 2SLS with controls.